

VADODARA HACKATHON 6.0

- Problem Statement ID - SIH25094
- Problem Statement Title - One-Stop Personalized Career & Education Advisor
- Theme - Smart Education
- PS Category - Software/Hardware
- Team ID - VH6_PIT_089
- Team Name – Anonymous bits



One-Stop Personalized Career & Education Advisor

Proposed Solution:

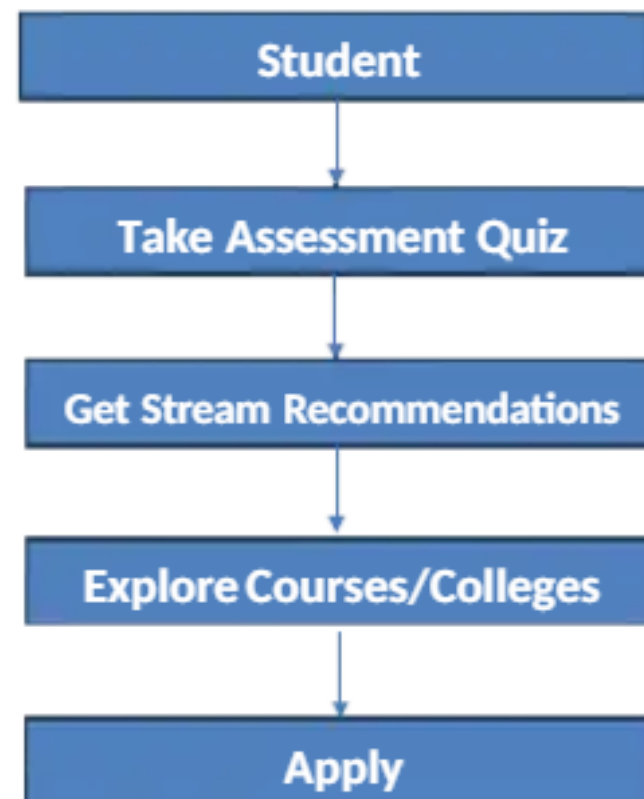
- Mobile + Web application for personalized student guidance.
- Aptitude quizzes to suggest streams & subjects.
- Course-to-Career path visualizations.
- Location-based government college directory.
- Timeline tracker for admissions & scholarships.
- AI-driven personalized user profiles.

Addresses the Problem By:

- Closing the critical awareness gap on career opportunities.
- Reducing post-Class 10/12 confusion and dropouts.
- Promoting local government college enrollment.
- Empowering students with data-driven decisions.

Innovation & Uniqueness:

- Guidance-first, AI-powered platform.
- Hyper-localized for Jammu & Kashmir.
- Designed for offline use in low-connectivity areas.
- Integrated stakeholder ecosystem.



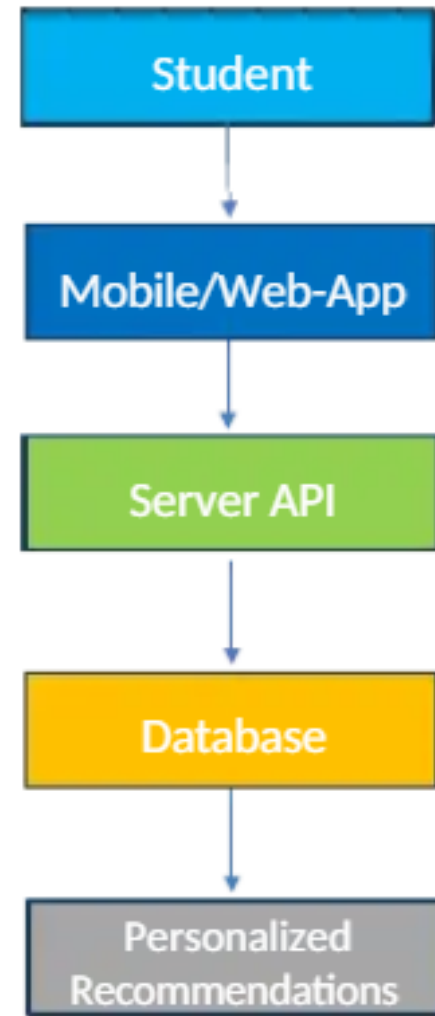
TECHNICAL APPROACH

Technologies Used:

- **Frontend:** React.js
- **Backend:** Node.js, Express.js
- **Database:** MongoDB
- **Deployment:** Netlify (Frontend), Railway (Backend)
- **Other Tools:** Figma (UI/UX), MongoDB Atlas (Cloud Database)

Implementation Methodology:

- Agile development process.
- Modular architecture for scalability.
- RESTful API for frontend-backend communication.
- Focus on minimal data usage and offline functionality.



FEASIBILITY AND VIABILITY

Feasibility Analysis:

- **High Technical Feasibility:** Built with popular, well-documented MERN stack.
- **Cost-Effective:** Uses free & open-source tools and affordable cloud services.
- **Scalable Architecture:** Designed to start small (pilot in J&K) and expand nationally.

Potential Challenges & Risks:

- Data collection on government colleges & courses.
- User trust and adoption in rural areas.
- Accuracy of the aptitude assessment algorithm.

Strategies to Overcome:

- Partner with Gov. education departments for official data.
- Work with school counselors & NGOs for grassroots outreach.
- Start with a simple rule-based system, improve with user feedback.

IMPACT AND BENEFITS

Impact on the target audience:

- Empowers J&K students to choose streams based on aptitude, not pressure.
- Reduces dropout rates post-Class 10/12 by providing clear career paths.
- Increases enrollment in local government colleges.
- Bridges information gap for underserved communities.

Key Benefits:

Social: Democratizes access to career guidance for all students

Economic: Builds a skilled workforce; prevents financial waste on wrong education

Educational: Strengthens the government higher education ecosystem

Strategic: Aligns with National Education Policy (NEP) 2020 goals

Collaborative: Fosters partnership between government, educators, and students

Students use App

Make Informed Choices

Enroll in Govt. Colleges

Reduce Dropout Rates

Build Skilled Workforce

RESEARCH AND REFERENCES

Research & Data Sources

- **National Education Policy (NEP) 2020:** Guidelines on vocational guidance and personalized learning.
- **AISHE Reports (All India Survey on Higher Education):** Data on enrollment trends in Arts, Science, Commerce streams.
- **National Career Service Portal (Govt. of India):** Studied for existing career guidance frameworks.
- **UGC Database:** Referenced for government college accreditation and program details.
- **More Information:** <https://youtu.be/YNZt9WnEjIM?feature=shared>

Future Research Directions

- Partnership with J&K Higher Education Department for real-time data integration.
- Collaboration with NCERT for psychometric assessment standardization.